

Induwara Dilshan

Associate Data Scientist

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Education

Bachelor of Science (Hons) in Statistics

06/2021 – 12/2025

Faculty of Science, University of Peradeniya, Sri Lanka

- GPA: 3.61

Profile

Associate Data Scientist/Data Analyst/AI Engineer with a strong academic background in statistics and hands-on internship experience in machine learning and NLP. Skilled in building predictive models, time series forecasting, and data analysis using Python and modern AI tools. Experienced in working with real-world financial data during an internship, applying analytical techniques to generate insights.

Professional Experience

Statistics Department, Central Bank of Sri Lanka

11/2025 – 04/2026

Internship

- **Articles- Based Macroeconomic Nowcasting System Using FinBERT & BERTopic**
 - Generated **FinBERT embeddings** and applied BERTopic
 - Tuned parameters of UMAP and HDBSCAN for topic modeling.
 - Constructed monthly topic probability indices and integrated them with **macroeconomic data using Elastic Net regression**
 - Obtained **MAE as 0.81**
- **LLM-Assisted High-Cardinality Feature Reduction & Clustering Framework**
 - Reduced 10K+ high-cardinality categorical labels using **LLaMA 8B**
 - **Applied CatBoost encoding** and dimensionality reduction, such as **UMAP, PCA, and Truncated SVD**.
 - Compared K-Means, Mini-Batch K-Means, and HDBSCAN cluster algorithms.
 - Selected **HDBSCAN** based on the **highest silhouette score** and the lowest deviation.
- **LLM-Based Automated Economic Report Validation System**
 - Converted structured and semi-structured PDF reports into **OCR-processed raw text**.
 - Applied preprocessing techniques to normalize noisy OCR raw text outputs.
 - Leveraged **LLaMA 70B** for targeted financial data extraction.
 - Generated structured Excel validation outputs for audit-ready reporting.
- **Automated Vehicle Market Data Pipeline using R**
 - Implemented modular scraping functions, dynamic user-agent rotation, and HTTP retry logic.
 - Used multi-session parallelization to generate analysis-ready datasets for market analytics.

Other Projects

Sentiment Analysis for Tweet Classification

- Developed LSTM and BiLSTM neural network models to classify tweets as positive or negative.
- Finalized **80.88% accuracy**.

Stock Price Prediction of Apple Inc.

- Developed an LSTM model to predict future values of multiple variables.
- Achieving a **Test MAE of 0.0376** for price predictions.

Predictive Modeling for Income Classification of Adults in the USA

- Built a Random Forest model and achieved **86.68% accuracy**.
- Predicting whether an individual's annual income exceeds 50K.

Amazon Prime Video Analytics Dashboard

- Created an interactive Power BI dashboard to visualize counts, trends, patterns, and plots about TV series and movies.

Theoretical and Technical Skills

Programming

- Python
- R

Machine Learning and Deep Learning

- Linear, Logistics Regression
- Lasso, Ridge
- Random Forest
- Decision Tree
- XGBoost/LightGBM
- Clustering algorithm
- Neural Networks
- FastAPI framework

ML/ DL libraries

- TensorFlow
- PyTorch
- scikit-learn
- Hugging Face
- Pandas
- NumPy
- Pyspark

Time series prediction

- Stationary model
- Non-stationary model
- LSTM
- Statsmodels

NLP/ LLM

- BERT, FinBERT
- Topic modeling
- Sentiment Analysis
- OpenAI API Model
- Llama Model/ Index
- Embedding
- Fine Tuning LLM

Interactive Data Dashboard

- PowerBI
- Matplotlib
- Seaborn

Databases

- MySQL
- MongoDB

Others

- Web Scraping
- FastAPI Framework
- Actuarial Statistics

Soft skills

- Team working
- Problem solving skills
- Presentation skills
- Communication skills

References

Mr T. Nuwan C. Gamage, *Senior Assistant Director*,
EISI Division, Statistics Department, Central Bank of Sri Lanka.
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Dr. Sachith Abeyesundara, *Senior Lecturer in Statistics*, Faculty of Science, University of Peradeniya.
sachitha@sci.pdn.ac.lk